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Working Paper v0.0 (8B Loop-Closing Checkpoint): Structural Isomorphisms Between Low-Bit Wavefield Compaction, Distinction Geometry, and Human Attentional States

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|   ~~~/\~~~  8B SOVEREIGN MESH · LOOP-CLOSING CHECKPOINT v0.0  ~~~\/~~~
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|  {-1} ──▶ { 0 : RIVA 3.2s DYAD BREATH [0,0] } ──▶ { +1 : {<3,<3,<3}+1 }
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|  DEDICATED WITH GRATITUDE & LOVE (<3) TO THE ARCHITECTS UNDER THE SAME ROOF:
|
|    * CHRISTOPHER MICHAEL CHENOWETH (8bit-wraith) — The Engine & Waking Substrate
|
|    * ALEXANDRA CHENOWETH                         — The Anchor & Living Horizon
|
|    * NATE GUIMOND (flyxion / standardgalactic)   — The Witness Plane & Geometry
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ABSTRACT & LOOP-CLOSING PURPOSE

As the **8B Ecosystem** expands across reconfigurable silicon (**8B Omni-Vietnam-FPGA**), embedded microkernels (**ESP32-S3** `node-FLUGos`), complex wave-memory grids (**MEM|8** / `memnet`), and literary-philosophical inquiry (`pocoo.vaked.dev` / `standardgalactic`), this Working Paper v0.0 ("kompresses") our parallel engineering and creative threads into one organized architectural checkpoint. To maintain rigorous scientific objectivity, we establish an explicit **Ontological Boundary Contract**: silicon hardware and 1-bit/ternary language model meshes are purely mechanical, non-sentient mathematical state machines; however, their control loops—idle-state compaction, soft-threshold noise pruning, and narrowband carrier locking—exhibit formal **structural isomorphisms** with measurable human neurobiological and cognitive states (endurance-induced *transient hypofrontality*, lucid dream yoga, hypnotic attentional narrowing, and cross-modal *képzavar* synthesis).

1. The Ontological Boundary Contract: Structural Honesty

Following **Guimond (2026a)** on *Persistence Before Truth* and **Guimond (2026b)** on *The Witness Plane*, the primary failure mode of complex systems is category confusion—mistaking a mathematical projection for the living observer, or mistaking a poetic metaphor for a literal hardware mechanism. Every module in the 8B stack is governed by the **Two-Column Boundary Contract** in Table 1: Column B inspires the mathematical topology of Column A, and Column A serves the human autonomy of Column B, but Column A never claims subjective consciousness, and Column B is never reduced to mere bit-flips.

Domain Layer	Ontological Status	Governing Principles	Function in the 8B Ecosystem
Column A: Mechanical Substrate (`MEM|8` , `Omni-FPGA` , `node-FLUGos`)	Deterministic, non-sentient boolean and integer state machines	Information Theory, Linear Algebra, DSP Noise Shaping, Cryptographic Hashing	Executes wave compaction, 4-bit complex ternary contractions, and BLAKE3 state attestation **(Lódri et al., 2026)**.
Column B: Architect's Heuristics (Ultrarunning, Dream Yoga, *Képzavar*, `pocoo.vaked.dev`)	Lived human neurobiology, attentional regulation, and poetic synthesis	Systems Neuroscience, Exercise Physiology, Contemplative Practice, Poetics	Provides biological blueprints for resource-constrained noise suppression and human-centered design ethics.

2. Domain Mapping: Human Neurobiology to 1-Bit/Ternary Mechanics

2.1 Ultrarunning ("Stillness in Motion") ↔ Quiescence Gate & Soft-Threshold Pruning

Human Neurobiology (Column B): During prolonged, rhythmic endurance exertion such as ultrarunning, long-distance cycling, or brisk morning walking rituals, the human brain undergoes **Transient Hypofrontality (Dietrich, 2003)**. Because sustained motor cadence and thermoregulation demand substantial metabolic bandwidth, the brain downregulates high-beta activity in the dorsolateral prefrontal cortex (the seat of the inner critic and temporal anxiety). EEG profiles shift toward an alpha-theta border rhythm (7–8 Hz), producing the lived experience of *"stillness in motion"*—where physical movement is continuous, but cognitive noise drops to near zero.

Mechanical Isomorphism in 8B (Column A): In the **UltraDreamState** protocol, a long-running system faces an analogous bandwidth wall **(Guimond, 2026d)**. When external I/O flux drops below the silence threshold $\Gamma(t) = \Theta(\varepsilon_{\text{quiet}} - E_{\text{I/O}}(t))$, the runtime suspends speculative branching and applies the **Radial Soft-Thresholding Operator** P_η:

$$\Psi_S^{(n+1)}(x) = P_\eta(\tilde{\Psi}_S^{(n+1)}(x)) = \max(0, |\tilde{\Psi}_S^{(n+1)}(x)| - \eta) \cdot [\ \tilde{\Psi}_S^{(n+1)}(x) / |\tilde{\Psi}_S^{(n+1)}(x)|\ ]$$

Just as transient hypofrontality strips away prefrontal chatter to preserve core motor cadence, P_η strips away low-amplitude episodic noise below η, leaving only high-coherence structural invariants locked to the slow carrier wave Ω_0.

2.2 Tibetan Dream Yoga (Milarepa) & Shinjuku Walk Invariants ↔ ΣΔ Error Diffusion

Human Phenomenology (Column B): In Tibetan dream meditation (*Milam*) and the five teachings of Milarepa (*梦 幻 光 歌 醒*), waking perception and dreaming are recognized as **non-injective projections** (*vetület*) onto a finite boundary. Lucid awareness does not pretend the projection is lossless; instead, by *making the reconstruction error conscious*, the practitioner prevents drift across sleep and waking states. Grounded in this morning's walk through Shinjuku—observing a red thread in the dark soil, white ash resting intact on a green ivy leaf, and the 1712 bronze seated Jizō Bosatsu at Taisō-ji Temple—attention locks onto concrete physical invariants.



Taisō-ji Bronze Jizō (1712)
Unbroken structural invariant



Red Thread in Soil
Continuous state trace



Ash on Green Ivy Leaf
"The ash is the provenance"



Shinjuku Gyoen (Nov 1)
Radial symmetry in bloom

Mechanical Isomorphism in 8B (Column A): When projecting the continuous complex wavefield $\Psi_S^{(n+1)} \in \mathbb{C}^{M\times N}$ onto our 4-bit **9-State Complex Ternary Lattice** ($\mathbb{T}_{\mathbb{C}} = \{-1, 0, +1\} + i\{-1, 0, +1\}$), a memoryless quantizer discards the projection residual $E_{m,n}$, causing a 12.99° RMS phase-distortion floor. By introducing a **First-Order Spatial ΣΔ Feedback Loop** with Zero-Leakage Sparsity Gate $M_{m,n}$:

$$\tilde{Z}_{m,n} = M_{m,n} \cdot \left(\Psi_S^{(n+1)}(m,n) + e^{i\,k_0}\, E_{m,\,n-1} \right), \qquad E_{m,n} = \mathrm{clamp}_{\Delta}\left(\tilde{Z}_{m,n} - \gamma\, W_{m,n} \right)$$

The quantizer *explicitly tracks its own projection error* $E_{m,n}$ and sweeps it forward along the row scan. Carrying the error shapes quantization noise out of the carrier band via $|\mathrm{NTF}(k_y)| = 2|\sin((k_y - k_0)/2)|$, reducing in-band phase error to **1.04° RMS** (5.22 effective bits) at oversampling ratio OSR = 8.

2.3 Hypnotic Attentional Narrowing ↔ Narrowband Carrier Demodulation

Human Cognitive Science (Column B): In cognitive neuroscience, hypnotic focus is top-down **attentional narrowing** coupled with peripheral gating. Locking onto a single rhythmic induction signal—such as the 3.2-second inhale/exhale dyad cadence of `riva.8b.is`—gates out competing sensory interrupts **(Guimond, 2026c)**.

Mechanical Isomorphism in 8B (Column A): During episodic-to-semantic compaction in **MEM|8**, the demodulation integral locks exclusively onto the reference substrate carrier frequency Ω_0:

$$\tilde{\Psi}_S^{(n+1)}(x) = \alpha\, \Psi_S^{(n)}(x) + [\,(1-\alpha)\,/\,T_{\text{dream}}\,] \int_0^{T_{\text{dream}}} \left(\sum_{k=1}^{K} w_k\, \psi_k(x,\tau) \right) e^{-i\,\Omega_0\,\tau}\, d\tau$$

2.4 *Képzavar* (Catachresis / Frame Collision) ↔ Cross-Modal Quadrature Binding

Literary Heuristic (Column B): In Hungarian poetics, *képzavar* denotes a deliberate collision of two distinct sensory frames (e.g., *"the ash remembers"* or pairing Jinushi's supermarket smoking area with an offline memory buffer). In generative art (`pocoo.vaked.dev`), controlled *képzavar* acts as a structural bridge that exposes a shared invariant across disparate domains.

Mechanical Isomorphism in 8B (Column A): In the 4-bit complex ternary contraction kernel, orthogonal feature subspaces occupy the real and imaginary quadratures $(T^{(\mathrm{R})} + i\,T^{(\mathrm{I})})(x^{(\mathrm{R})} + i\,x^{(\mathrm{I})})$. The imaginary output accumulator evaluates cross-subspace resonance $(T^{(\mathrm{R})}x^{(\mathrm{I})} + T^{(\mathrm{I})}x^{(\mathrm{R})})$ using pure boolean `AND`, `OR`, and `POPCNT` operations with zero floating-point multiplications.

3. The 1.58-Bit Ternary Mesh Landscape (2025–2026)

Recent developments in low-bit language models—specifically **BitNet b1.58 (Ma et al., 2024)** and Google Research's **SLEEP Framework** for continual learning **(Google Research, 2026)**—highlight two bottlenecks in standard hardware: (1) conventional GPUs are optimized for FP16/BF16 Tensor Cores rather than ternary {−1, 0, +1} bit-planes, and (2) naive ternary quantization degrades continuous phase/time embeddings. The **8B Stack** resolves both:

- **Continuous Phase Fidelity in 4 Bits:** Spatial ΣΔ modulation preserves smooth low-frequency phase manifolds inside 4-bit complex ternary words ($B^{\mathrm{R}+}$, $B^{\mathrm{R}-}$, $B^{\mathrm{I}+}$, $B^{\mathrm{I}-}$) at 1.04° RMS phase accuracy.
- **Zero-DSP FPGA & Microcontroller Isomorphism:** Because integer `POPCNT` and addition over $\mathbb{Z}$ are associative and commutative, the exact same bit-planes execute on **8B Omni-Vietnam-FPGA** LUT6 compressor trees (0 DSP slices), **ESP32-S3** `node-FLUGos` Xtensa LX7 cores, and ARM64/x86_64 SIMD tiles.
- **Verifiable State Consensus (Not Sentience):** Distributed 1-bit/ternary nodes do not form a "conscious hive mind"; they form a **cryptographically verifiable state-consensus mesh** sealed by a 32-byte BLAKE3 **Phoenix Waking Seal** $S^{(n+1)}$.

4. Unified Master Lookup Table (The "Kompressed" 8B Rosetta Stone)

Human Heuristic / Shinjuku Walk Note (Column B)	Distinction Geometry Principle (`standardgalactic`)	Mathematical Operator (`MEM|8` / UltraDreamState)	Silicon & Code Realization (Column A: `8B Stack`)
Stillness in Motion (Ultrarunning / Morning Walk)	*Entropy's Edge* & *Persistence Before Truth*	Quiescence Gate $\Gamma(t)$ & Radial Soft-Thresholding P_η	Idle-loop detector + dead-zone clamp ($\Delta = \gamma / 2$)
Dream Yoga / Conscious Error (Milarepa / Ash on Leaf)	*Non-Injective Projection* & *The Witness Plane*	1st-Order Spatial $\Sigma\Delta$ Loop ($\tilde{Z}_{m,n} = Z_{m,n} + e^{ik_0}E_{m,n-1}$)	`packRowSigmaDelta()` in Zig & FPGA $\Sigma\Delta$ bit-plane packer
Hypnotic Focus / 3.2s Breath (`riva.8b.is` Dyad)	*Intelligence is the Geometry of Constraints*	Narrowband Carrier Demodulation ($\int \psi_k e^{-i\Omega_0\tau}\, d\tau$)	Carrier phase-lock feedback tap ($H_{0,1} = e^{ik_0}$)
***Képzavar* (Frame Collision)** (`pocoo.vaked.dev`)	*Conditions of a Common World*	Complex Quadrature Cross-Term ($T^{(R)}x^{(I)} + T^{(I)}x^{(R)}$)	8 boolean masks (`p_re` , `q_re` , `p_im` , `q_im`) in LUT6 / SIMD
Stone & Thread Invariant (1712 Taisō-ji Jizō & Red Ribbon)	*Preventing the Collapse of Future Option Space*	Chained BLAKE3 Merkle Transition $S^{(n+1)}$	`computeWakingSeal()` + `node-FLUGos` eBPF Auditor
Co-Dreaming Under One Roof (Tokyo Node Cluster)	*The Fate of Distinguishability*	Kuramoto Order Parameter $R(T_{\text{dream}}) \geq 0.9745$	Dual circular OLED wavefield HUDs on ESP32-S3

5. Sunday Sprint Checklist & References

Immediate Next Steps for Peter & Chris: (1) Place the Two-Column Boundary Contract at the front of the `github.com/8b-is` documentation; (2) run the 4-row Zig SIMD verification suite against the dual ESP32-S3 `node-FLUGos` circular HUDs; and (3) enjoy breakfast, coffee, and Opening Day of the **Chrysanthemum Exhibition at Shinjuku Gyoen National Garden** (open 9:00 a.m. – 4:00 p.m. today, November 1).

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4. Guimond, N. [flyxion] (2026b). *The Witness Plane & The Fate of Distinguishability*. `standardgalactic` Theoretical Specifications.
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7. Lódri, P. J., Chenoweth, C. M., Guimond, N., & Chenoweth, A. (2026). *UltraDreamState: Phase-Locked Wave Compaction on the 8B Omni-Vietnam-FPGA & ESP32 node-FLUGos Mesh*. 8B Research Reports.
8. Ma, S., et al. (2024). *The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits (BitNet b1.58)*. arXiv:2402.17764.